

NEXUS DESK TRADING TERMINAL

Technical Architecture, Multi-Agent Workflow & In-Context Learning Engine

1. Executive Overview & Design Principles

Nexus Desk is an autonomous institutional paper-trading terminal built with a multi-agent hierarchy and a fail-closed risk management engine. Rather than operating as an unobservable, black-box trading system, the terminal enforces explicit boundaries between Signal Detection, Pre-Flight Risk Filtering, Multi-Agent Veto Gates, and Execution Monitoring.

- **Fail-Closed Architecture:** Trading halts immediately if connectivity breaks, a circuit breaker trips, or the manual Kill Switch is engaged.
- **Human-in-the-Loop Supervision:** Three execution modes (Manual, Semi-Auto, Auto Within Limits) allow operators to maintain direct veto authority over strategy recommendations.
- **Continuous In-Context Adaptation:** Closed trades automatically undergo post-mortem autopsies that update a persistent Memory Bank to reject low-probability setups in future iterations.

2. Multi-Agent Hierarchy & Roles

The terminal coordinates five specialized algorithmic agents, each responsible for an isolated step in the trade decision pipeline:

- **Market Radar Scanner:** Continuously ingests order book ticks, monitoring price velocity, 24-hour volume surges, and RSI divergence across pairs (BTC, ETH, SOL, JUP, etc.).
- **Strategy Agent (Breakout-v2.0):** Evaluates trend breakouts above dynamic resistance levels with volume threshold confirmation. Automatically computes calibrated Entry, Stop-Loss, and Take-Profit price levels.
- **Pre-Flight Risk Engine:** Evaluates candidate proposals against capital constraints, ensuring gross open positions do not exceed 5, individual ticket sizing respects max equity percentage, and market spread remains acceptable.
- **Commander Supervisor:** Maintains high-level oversight of gross desk exposure, win/loss streaks, and market volatility regimes. Authorizes autonomous self-approval and generates the executive briefing.
- **Autopsy & Reflection Agent:** Inspects every closed position immediately upon exit, categorizing whether profit or loss was driven by trend continuation, false breakout, or chop, and formulating persistent lessons.

3. Step-by-Step Execution Lifecycle

Step 1: Signal Identification & Radar Scoring

The Market Radar analyzes high-frequency tick data. When an asset crosses key momentum thresholds and breaks its 20-period moving average with elevated relative volume, a trade candidate is flagged.

Step 2: Proposal Formulation & Risk Verification

The Strategy Agent computes position sizing using ATR (Average True Range) stop-loss positioning and a 2:1 reward-to-risk ratio. The proposal enters the Queue with status PENDING_APPROVAL and an agent confidence score (e.g., 88%).

Step 3: Approval Gate & Clearance Enforcement

The system verifies operator clearance and active decision mode. In MANUAL mode, trades require Commander or Trader approval. In AUTO_WITHIN_LIMITS, the Commander agent cross-references the proposal against recent memory lessons and executes automatically if no veto conditions are triggered.

Step 4: Active Book Monitoring & Order Execution

Approved orders enter the active Book. The execution loop recomputes unrealized PnL every 2 seconds. Automated stop-losses protect downside capital, take-profit limits capture returns at targets, and operators can trigger immediate manual market exits.

4. Current Training & Learning Methodology

In live quant trading, neural network weights are not continuously fine-tuned directly on noisy financial price ticks to prevent catastrophic forgetting and overfitting. Nexus Desk utilizes a sophisticated hybrid approach combining Mathematical Walk-Forward Optimization with LLM In-Context Memory Reflexion:

A. Reflexion & Memory Bank (The Post-Mortem Loop)

- **Trigger Event:** Every closed position triggers an automated autopsy function.
- **Diagnostic Extraction:** The system identifies entry conditions, market regime (trending, consolidating, choppy), and the primary reason for success or stop-out.
- **Distilled Heuristic:** The post-mortem translates the loss into an explicit heuristic (e.g., 'Do not long SOL when BTC 15m correlation is divergent and volume is declining').
- **Prompt Pre-Conditioning:** Future candidate proposals are matched against active memory heuristics. If a candidate matches a prior failure condition, confidence is degraded or auto-vetoed.

B. Walk-Forward Lab (Quantitative Model Selection)

- **Train / Test Slices:** The Lab tests model configurations across rolling historical windows (e.g., 30-day in-sample training, 7-day out-of-sample testing).
- **Champion vs. Challenger:** The active 'Champion' model is benchmarked against candidate models on Sharpe Ratio, Max Drawdown, Profit Factor, and Win Rate.
- **Promotion Gate:** A challenger model can only be promoted to active production trading if it statistically outperforms the Champion across out-of-sample test splits.

5. Security, RBAC & Immutable Audit Ledger

Nexus Desk integrates Firebase Authentication and Firestore Security Rules to enforce strict operational clearance:

- **Commander Clearance:** Full operational authority. Can toggle the emergency Kill Switch, activate Autonomous Self-Approval, and promote models in the Lab.
- **Desk Trader Clearance:** Execution authority. Can approve or veto queued tickets and execute manual position exits. Prohibited from enabling Autonomous mode.
- **Auditor Clearance:** Read-only compliance authority. Can observe the trading floor, review active book positions, and inspect real-time audit trails without trade execution permissions.
- **Immutable Audit Ledger:** All critical actions (sign-in, role changes, kill switch toggles, order approvals, manual exits) are permanently written to Cloud Firestore under strict security rules forbidding updates or deletions.